

1) a) $f_n(x) = n I_{[0, \frac{1}{n}]}(x)$ $f(x) = 0$.

(i) As $n \rightarrow \infty$, $[0, \frac{1}{n}] \rightarrow \{0\}$.

$$I_{[0, \frac{1}{n}]} \rightarrow 0 \text{ a.e.}(m) \text{ b/c } m(\{0\}) = 0$$

$$\text{So, } n I_{[0, \frac{1}{n}]} \rightarrow 0 \text{ a.e.}(m) \text{ b/c } m(\{0\}) = 0.$$

$$\text{So } f_n(x) = n I_{[0, \frac{1}{n}]}(x) \rightarrow 0 = f(x) \text{ a.e.}(m). \quad \checkmark$$

(ii) $a_n(t) = \int_{\{H_n \geq t\}} |f_n| dm$ $\sup_{n \geq 1} a_n(t) \rightarrow 0$ as $t \rightarrow \infty$?

Fix t . ^{still large} Then there are $n > t$, for $n > t$, $a_n(t) =$

$$a_n(t) = \begin{cases} 0 & n < t \\ n(\frac{1}{n}) = 1 & n \geq t \end{cases}$$

So, for $n < t$, $a_n(t) = 0$, but for $n \geq t$, $a_n(t) = 1$.

So, $\sup a_n(t) = 1 \neq 0$, Not UI. ✓

| a cont |

$$\text{iii) } \int |f| d\mu < \infty \quad \int |f| d\mu < \infty.$$

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu \rightarrow \lim_{n \rightarrow \infty} \int |f_n| d\mu =$$

$$\text{As } n \rightarrow \infty, f_n = n \cdot \mathbb{I}_{[0, \frac{1}{n})}$$

$$\lim_{n \rightarrow \infty} \int |n \cdot \mathbb{I}_{[0, \frac{1}{n})}| d\mu = \lim_{n \rightarrow \infty} (n \cdot \frac{1}{n}) = \lim_{n \rightarrow \infty} (1) = 1 \neq 0.$$

$$\text{So, } f_n \not\rightarrow f.$$

| b | Using Scheffe's Theorem;

$\int f d\mu \rightarrow \int f d\mu$, $\int f d\mu$ is finite,

$$\boxed{p=1}?$$

gen. p? -2

Add limits
Doesn't change
math...

$$| c | \lim_{n \rightarrow \infty} \int |f_n - f|^p d\mu \rightarrow \lim_{n \rightarrow \infty} \int |f_n - f|^q d\mu \quad \text{for } q < p$$

Use Holder's Inequality. $f = |f_n - f|^p \quad q = 1$

$$\int |f_n - f|^q d\mu \leq \left(\int |f_n - f|^p d\mu \right)^{\frac{1}{p}} \underbrace{\left(\int 1 d\mu \right)^{\frac{1}{q}}}_{=1 \cdot \forall q} = \left(\int |f_n - f|^p d\mu \right)^{\frac{1}{p}}$$

Let $p^* = \frac{p}{q}$ then $q^* = \frac{p/q}{p/q - 1}$ Does not matter, then

$$\left(\int |f_n - f|^p d\mu \right)^{\frac{q}{p}} = \left(\int |f_n - f|^p d\mu \right)^{\frac{q}{p}}. \quad \text{Then } \int |f_n - f|^q d\mu \leq \left(\int |f_n - f|^p d\mu \right)^{\frac{q}{p}}$$

Since $\int |f_n - f|^p d\mu$ is finite, $\int |f_n - f|^q d\mu$ is finite, so $f_n \rightarrow f$ in L^q

-1

2

a) $M(t) = E(e^{tx})$

Show $P(X > a) \leq \inf_{t \geq 0} e^{-ta} M(t)$. $a > 0$.

$P(X > a) \leq \inf \left(\frac{e^{tx}}{e^{ta}} \right)$

Can remember
inequality
it looked like
that

$P(X > a) = \int f_X I_{[a, \infty)} d\mu$ f_X is pdf for X .

$\int f_X I_{[a, \infty)} d\mu \leq \inf \left(\frac{e^{tx}}{e^{ta}} \right)$

by _____ inequality?

?

-6.

b) If $X \geq 0$ is R.V. w/ $Var(X) < \infty$. $P(X > 0) \geq \frac{[E(X)]^2}{E(X^2)}$

$E(X) \leq E(|\sqrt{X}|^2)^{1/2} E(|\sqrt{X}|^2)^{1/2}$

$E(X)^2 \leq E(X^2) P(X > 0)$

$E(X) \leq E(X)^{1/2} E(X)^{1/2}$

$E(X^2) \leq E(X)^{1/2} E(X)^{1/2}$

$E(|X|^2) \leq E(|X|^2)^{1/2} E(|X|^2)^{1/2}$

$E(X) \leq E(X)^{1/2}$

$E(X)^2 \leq E(X^2)$

~~Wrong sol~~

X ~~OP~~

-6.

$\sqrt{E(X^2)} \geq E(X)$

$E(X)^2 \leq E(X^2)$

3

a) For $|f_n| \leq g$, $\int |g| d\mu < \infty$, and $f_n \rightarrow f$ a.e. (μ),

$f_n \in L^1$, $\int f_n d\mu \rightarrow \int f d\mu$, and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

b) Subsequence

For some subsequence $\{f_{n_k}\}_{k \geq 1}$, since $f_n \xrightarrow{m} f$,

$f_{n_k} \rightarrow f$ a.e. (μ).

We can plug in and use that subsequence in the DCT and we are back to $f_{n_k} \rightarrow f$ a.e. (μ) which is our condition.

?

-8

4

a) $\mu_3(\emptyset) = 0$
 $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0 + 0 = 0.$

$$\underline{\mu_3(A \cup B) = \mu_3(A) + \mu_3(B)}$$

$$\begin{aligned} \mu_3(A \cup B) &= \mu_1(A \cup B) + \mu_2(A \cup B) = \mu_1(A) + \mu_1(B) + \mu_2(A) + \mu_2(B) \\ &= (\mu_1(A) + \mu_2(A)) + (\mu_1(B) + \mu_2(B)) = \mu_3(A) + \mu_3(B). \end{aligned}$$

$$\underline{\mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_3(A_n)}$$

$$\begin{aligned} \mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) &= \mu_1\left(\bigcup_{n=1}^{\infty} A_n\right) + \mu_2\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_1(A_n) + \sum_{n=1}^{\infty} \mu_2(A_n) \\ &= \sum_{n=1}^{\infty} (\mu_1(A_n) + \mu_2(A_n)) = \sum_{n=1}^{\infty} \mu_3(A_n). \end{aligned}$$

So, μ_3 is a measure.

b) $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.

Take a set $A \in \mathcal{F}$ s.t. $\nu(A) = 0$. Then $\mu_1(A) = 0$ and $\mu_2(A) = 0$.

Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$. Then, $\mu_3 = \mu_1 + \mu_2$, so $\mu_3(A) = 0 + 0 = 0$.

So $\forall$ set A s.t. $\nu(A) = 0$, $\mu_3(A) = 0$. $\mu_3 \ll \nu$. ✓

~~Q.E.D.~~

4c

$$\mu_1 = \int \frac{d\mu_1}{dv} dv$$

$$\mu_2 = \int \frac{d\mu_2}{dv} dv$$

$$\mu_3 = \mu_1 + \mu_2 = \int \frac{d\mu_1}{dv} dv + \int \frac{d\mu_2}{dv} dv$$

$$= \int \left(\frac{d\mu_1}{dv} + \frac{d\mu_2}{dv} \right) dv.$$

Then, since $\mu_3 \ll v$,

$$\frac{d\mu_3}{dv} = \frac{d\mu_1}{dv} + \frac{d\mu_2}{dv}.$$

D $\mu \ll v$ $\mu \perp v \rightarrow \mu$ is zero measure.

For some $A \in \mathcal{F}$, $v(A) = 0$ and $\mu(A^c) = 0$ b/c $\mu \perp v$.

We also know that since $v(A) = 0$, $\mu(A) = 0$ b/c $\mu \ll v$.

So, $\mu(A^c) = 0$ and $\mu(A) = 0$, so $\mu(B) = 0$ for any set $B \in \mathcal{F}$.

So, μ is the zero measure.

4e] $\mu_1 = \text{lognormal}(5,6)$ $\mu_2 = \text{Geometric}(.3)$
 $(0,1,\dots)$
 $\nu = \text{Poisson}(8), (0,1,2,\dots)$

$\mu_{3a} = \text{Geometric}(.3)$

$\mu_{3s} = \text{lognormal}(5,6)$

$\mu_{3a} \ll \nu$

ν is ~~non~~ nonzero only on $\{0,1,2,\dots\}$.

So ν is zero only on $\mathbb{R} \setminus \{0,1,2,3,\dots\}$.

Take $A = \mathbb{R} \setminus \{0,1,2,3,\dots\}$.

$\nu(A) = 0$ and since μ_{3a} is only nonzero on $\{0,1,2,3,\dots\}$

$\mu_{3a}(A) = 0$ as well.

Since A is entire set where $\nu(A) = 0$, this proves dominance.

$\mu_{3s} \perp \nu$

Take $A = \{0,1,2,3,\dots\}$

$\nu(A^c) = 0$ and $\mu_{3s}(A) = 0$

or
 No nonnegative integers
 in A^c .

← Since A is countable + μ_{3s} is continuous,

Continued on
 next page...

4e continued

$$\mu_{3a} = \int_A f_{\text{geom}} d\mathbf{c} \quad \text{where } \mathbf{c} \text{ is counting measure on } \{0, 1, 2, \dots\}.$$

$$= \int_A \frac{f_{\text{geom}}}{f_{\text{poisson}}} d\mathbf{c}$$

$$= \int_A \frac{f_{\text{geom}}}{f_{\text{poisson}}} d\nu$$

So, by part (ii) of decomposition theorem,

$$\frac{d\mu_{3a}}{d\nu} = \frac{f_{\text{geom}}}{f_{\text{poisson}}} = \frac{.3 (.7)^{x-1}}{\frac{e^{-8} 8^x}{x!}} = \frac{x! (.3) (.7)^{x-1}}{e^{-8} 8^x}$$

This is rational $\frac{3}{e}$

$$\frac{d\mu_{3a}}{d\nu} = \frac{d\mu_{3a}}{d\mathbf{c}} \left(\frac{d\mathbf{c}}{d\nu} \right) = \left(\frac{d\mu_{3a}}{d\mathbf{c}} \right) \left(\frac{d\nu}{d\mathbf{c}} \right)^{-1} \quad \text{for } \mathbf{c} \text{ is counting on } \{0, 1, 2, \dots\}.$$